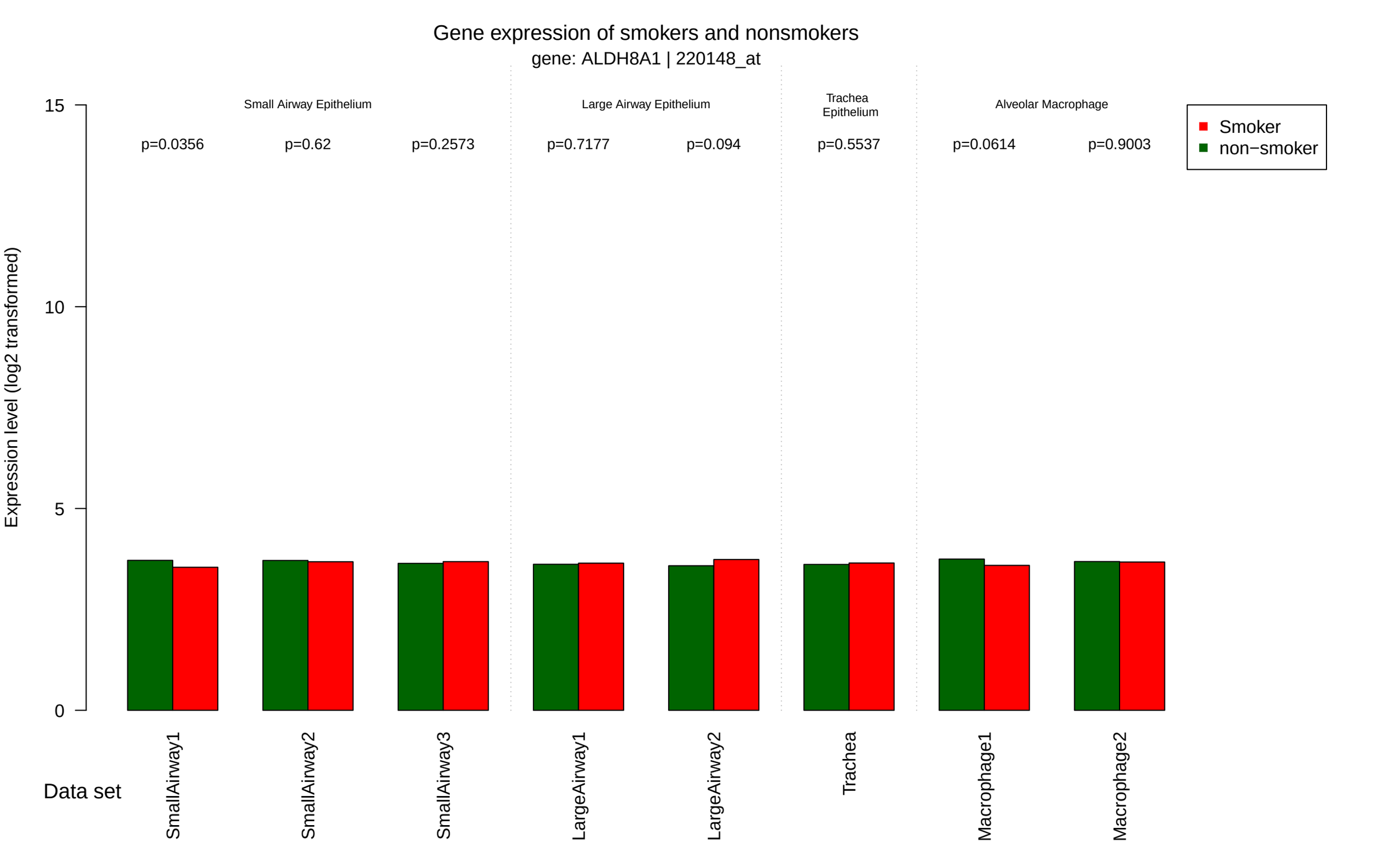
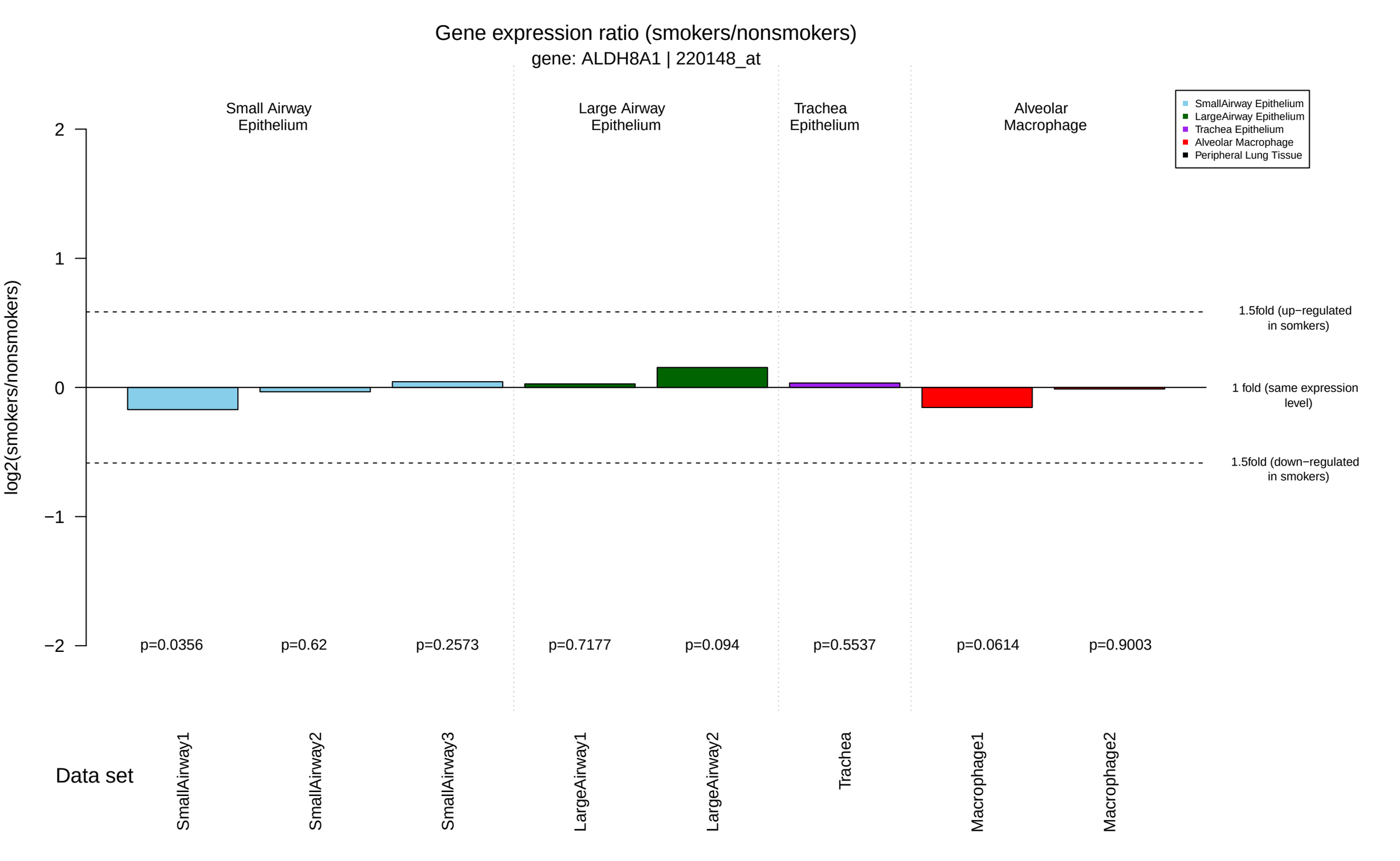
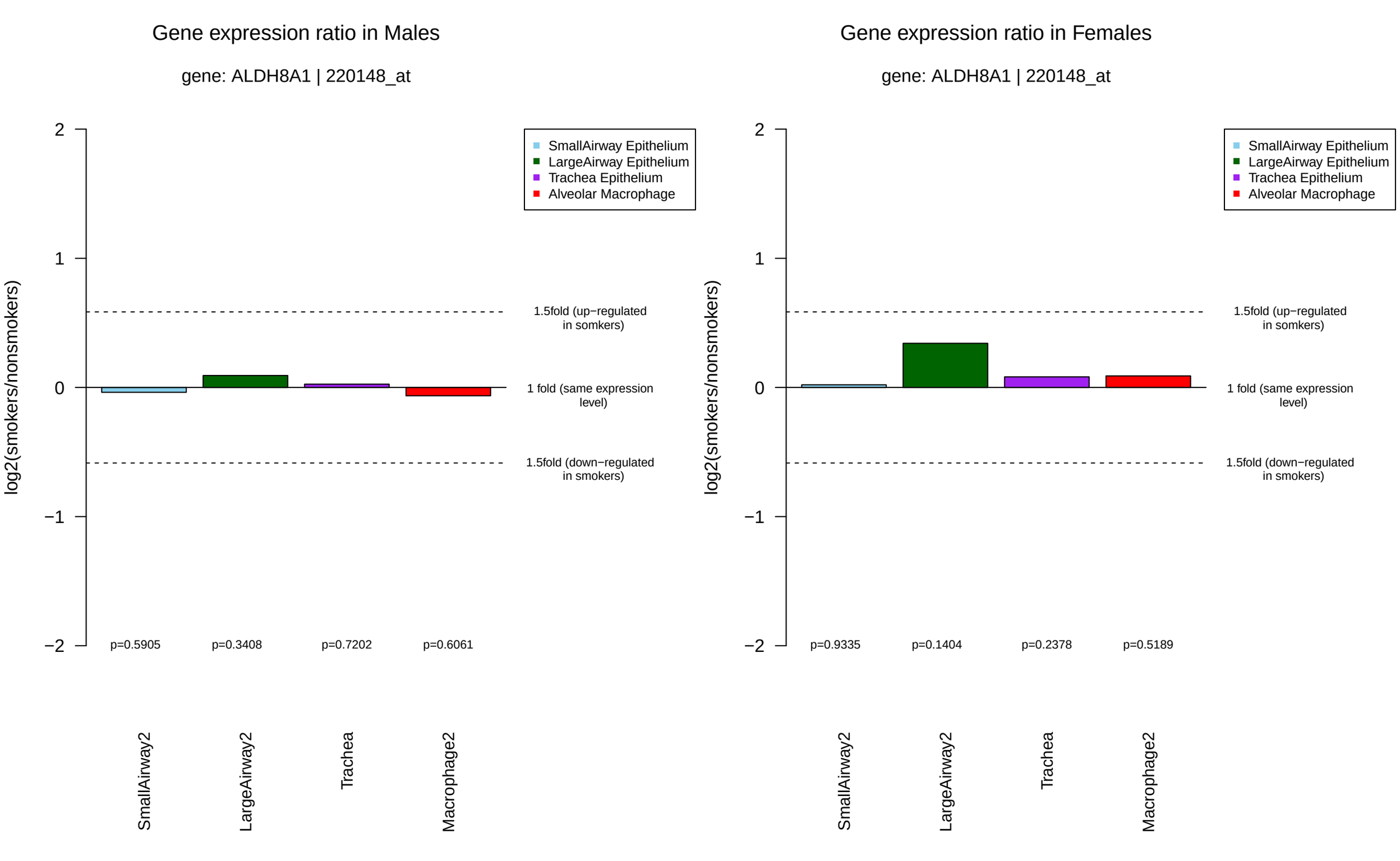
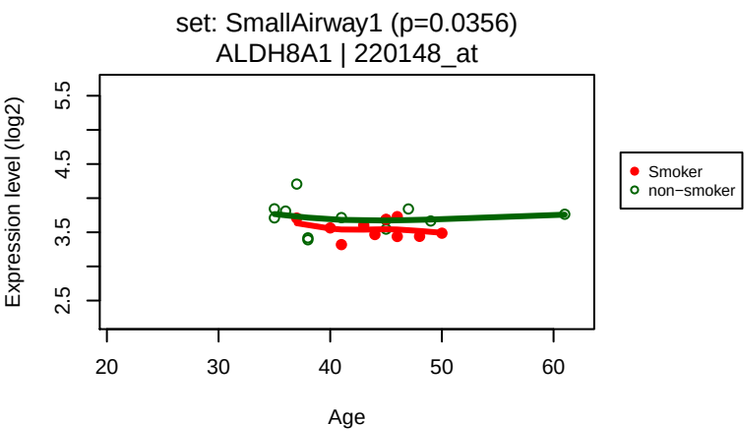
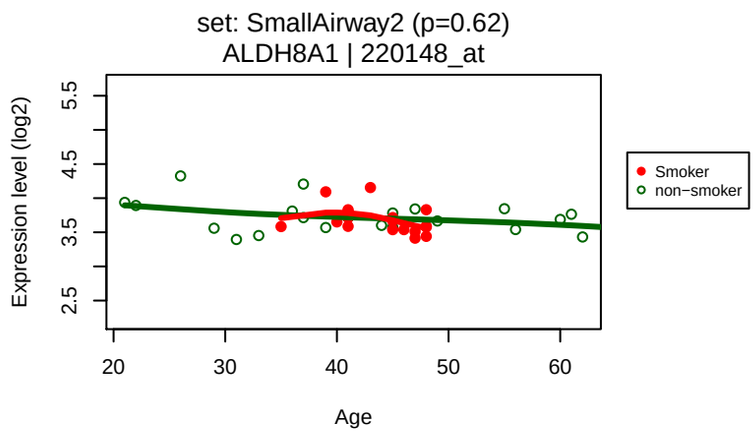
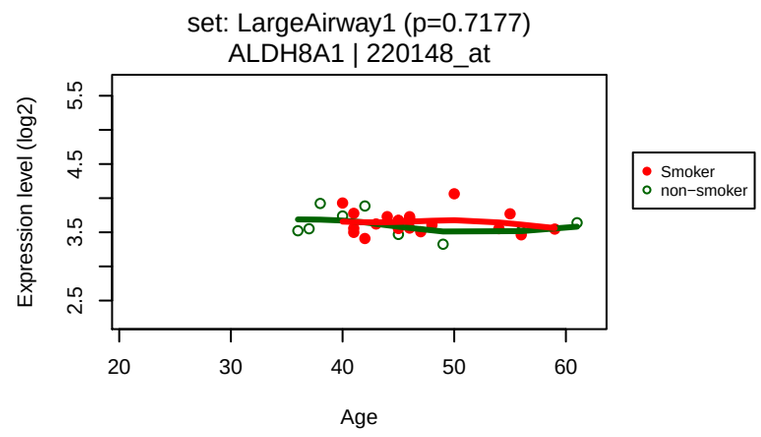

Gene expression of smokers and nonsmokers
gene: ALDH8A1 | 220148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0356
p=0.62
p=0.2573
p=0.7177
p=0.094
p=0.5537
p=0.0614
p=0.9003
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

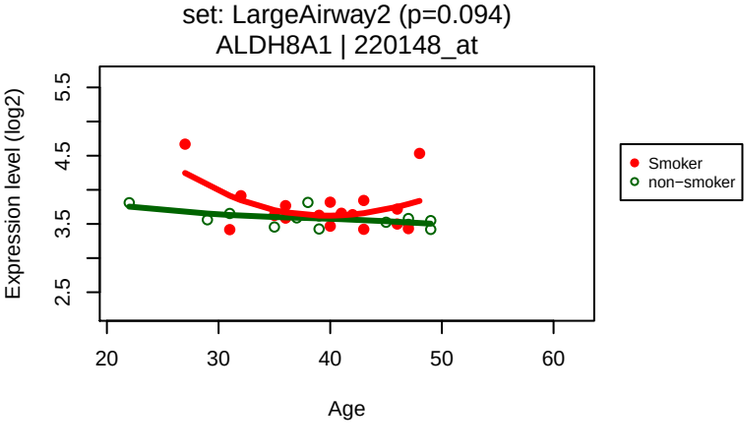

Gene expression ratio (smokers/nonsmokers)
gene: ALDH8A1 | 220148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0356
p=0.62
p=0.2573
p=0.7177
p=0.094
p=0.5537
p=0.0614
p=0.9003
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

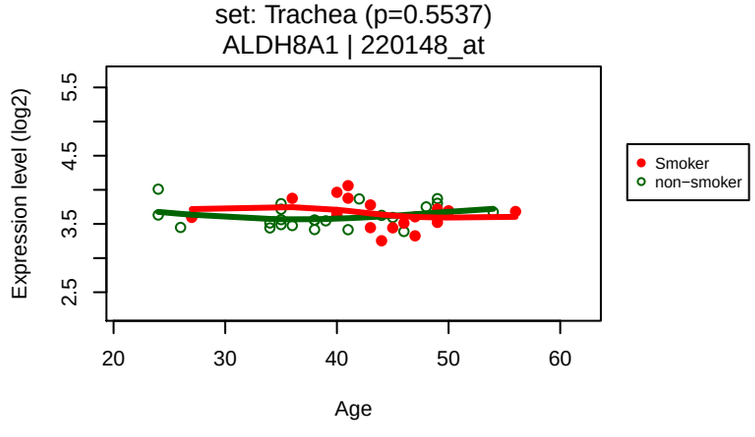

Gene expression ratio in Males
gene: ALDH8A1 | 220148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5905
p=0.3408
p=0.7202
p=0.6061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH8A1 | 220148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9335
p=0.1404
p=0.2378
p=0.5189
SmallAirway2
LargeAirway2
Trachea
Macrophage2

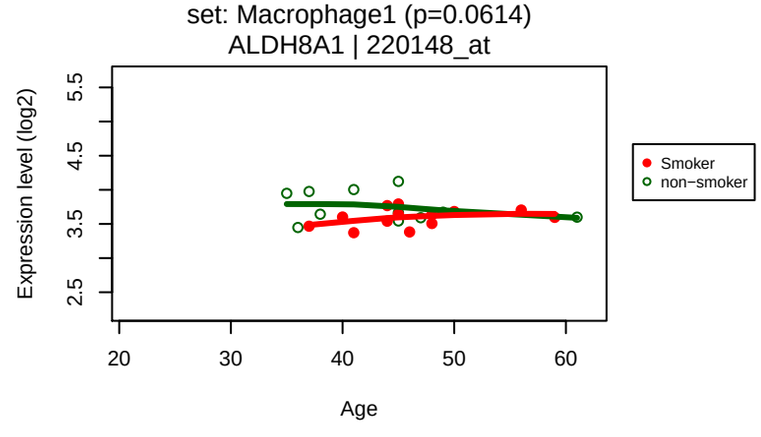

set: SmallAirway1 (p=0.0356)
ALDH8A1 | 220148_at
Expression level (log2)
Age
Smoker
non−smoker

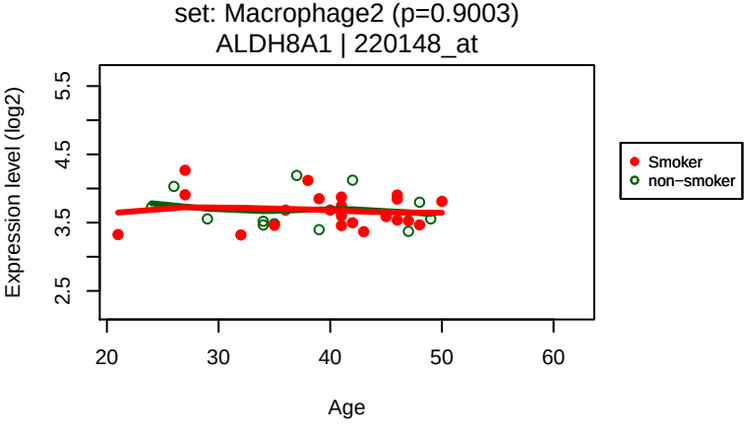

set: SmallAirway2 (p=0.62)
ALDH8A1 | 220148_at
Expression level (log2)
Age
Smoker
non−smoker

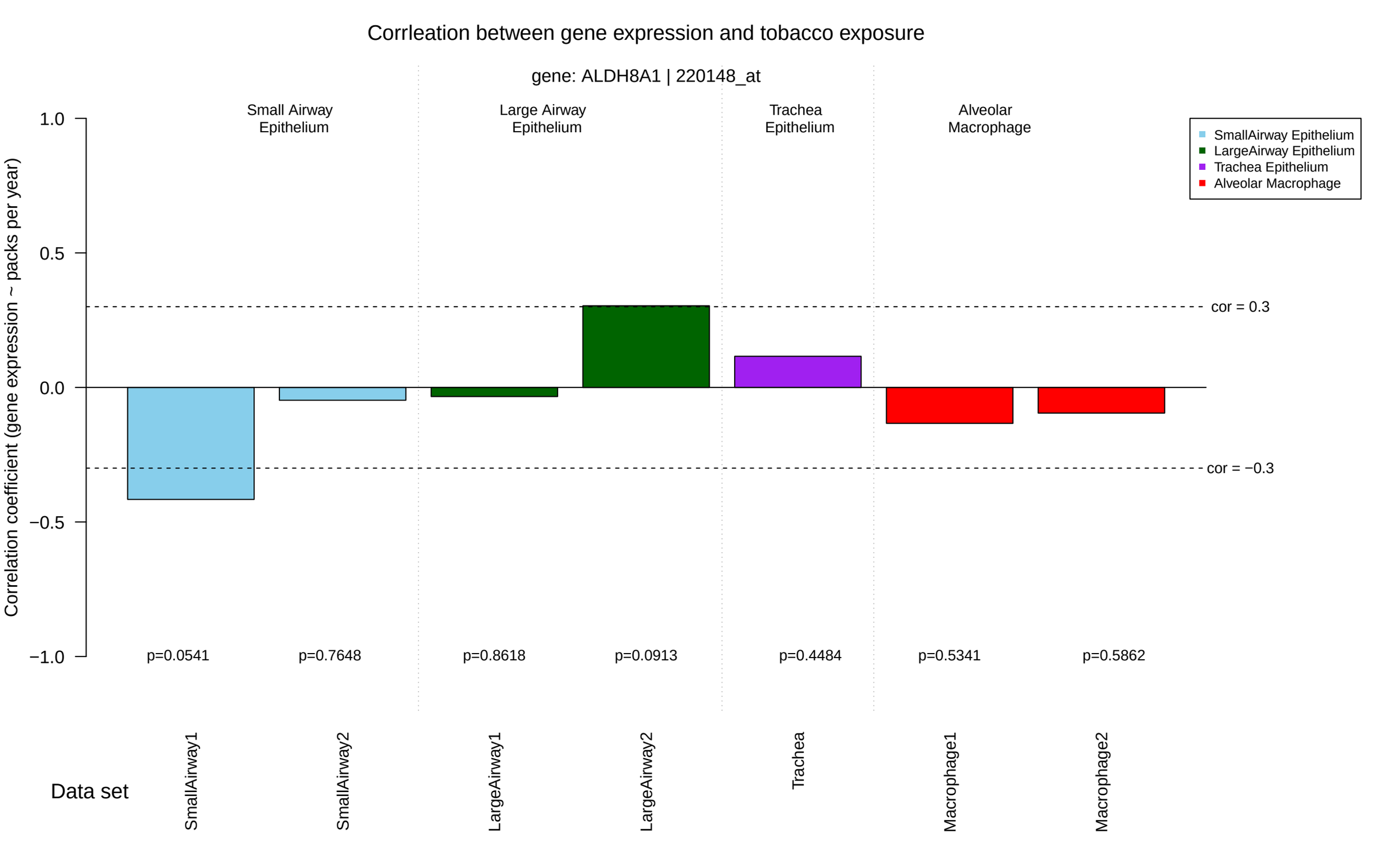

set: LargeAirway1 (p=0.7177)
ALDH8A1 | 220148_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.094)
ALDH8A1 | 220148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5537)
ALDH8A1 | 220148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0614)
ALDH8A1 | 220148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9003)
ALDH8A1 | 220148_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH8A1 | 220148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0541
p=0.7648
p=0.8618
p=0.0913
p=0.4484
p=0.5341
p=0.5862
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2